

SAKTHIVEL NATHAN

SDET — Automation Test Engineer (4+ Years Experience)

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PROFESSIONAL SUMMARY

SDET with 4+ years of experience building scalable UI automation frameworks using Playwright and TypeScript. Designed automation solutions for Business Process Automation platforms, reducing flaky tests through custom locator strategies and JSON-driven workflow generation. Experienced in AI testing using DeepEval, CI/CD integration, and improving automation reliability for rapidly evolving applications.

TECHNICAL SKILLS

- **Programming Languages:** TypeScript, JavaScript, SQL
- **Automation Frameworks:** Playwright, Page Object Model (POM), BDD, Data Driven Framework
- **API Testing:** REST API Testing, Postman
- **LLM Evaluation:** DeepEval
- **CI/CD:** GitLab
- **Testing Methodologies & Tools:** Functional Testing, Regression Testing, Smoke Testing, Sanity Testing, Cross Browser Testing, JIRA, Azure DevOps

PROFESSIONAL EXPERIENCE

VIT Infotech <i>Senior Software Engineer (SDET)</i>	Mar 2025 – Present <i>Bengaluru, India</i>
• Built a scalable Playwright and TypeScript framework supporting 600+ UI test cases for a node-based Business Process Automation platform. • Developed a modular Page Object Model architecture with Playwright session states to validate role-based permissions across 5+ application screens. • Implemented a fallback locator strategy to handle dynamic UI changes, reducing test maintenance effort by 80% and improving test stability. • Developed a JSON-driven dynamic workflow generator to minimize duplicate test logic and reduce creation time for complex workflows by 70%. • Integrated DeepEval-based LLM evaluation in automation pipelines to validate 25+ AI agent scenarios using rag metrics. • Implemented multi-layer validation covering workflow execution status, node outputs, JSON payloads, dynamic data changes, and SQL database records. • Optimized Playwright execution time from 6 hours to 2 hours by configuring parallel multi-worker execution in GitLab CI.	
Sword Group <i>Software Tester</i>	Oct 2022 – Mar 2025 <i>Chennai, India</i>
• Increased automated regression coverage by 40% in multiple modules using Playwright and TypeScript. • Created and maintained 150+ testcases also providing timely reports to business analysts and engineering teams. • Validated backend REST APIs using Postman and Playwright Request API to verify HTTP response. • Tracked and managed the defect lifecycle in 20+ Agile sprints using JIRA while collaborating with cross-functional teams. • Performance testing of cross-browser functional, smoke, and regression using Playwright in Chrome, Firefox, and Safari environments.	

EDUCATION

Karpagam Academy of Higher Education Master of Engineering (ME) – 8.5 CGPA	Graduated: 2022 <i>Coimbatore, India</i>
Karpagam College of Engineering Bachelor of Engineering (BE) – 8.0 CGPA	Graduated: 2020 <i>Coimbatore, India</i>